```
clear; clc; close all;

%% 1. Set MIT-BIH Path
mitdb_path = 'D:\UNIVERSITY\6th Semester\SS Lab\project\mit-bih-arrhythmia-
database-1.0.0';
record_name = '100';

%% 2. Robust MIT-BIH Reader
function [ecg, Fs] = local_read_mitbih(record_name, mitdb_path)
    % Read .dat file (format 212)
    dat_file = fullfile(mitdb_path, [record_name '.dat']);
    fid = fopen(dat_file, 'r');
    data = fread(fid, [2, Inf], 'int16')';
    fclose(fid);
    ecg = data(:,1);

    header_file = fullfile(mitdb_path, [record_name '.hea']);
    fid = fopen(header_file, 'r');
    header_line = fgetl(fid);
    fclose(fid);

    header_parts = strsplit(header_line);
    Fs = str2double(header_parts{3});

    if isempty(Fs) || Fs < 250
        error('Invalid Fs=%d Hz in header. Expected ≥360 Hz.', Fs);
    end
end

%% 3. Load Data with Validation
[ecg, Fs] = local_read_mitbih(record_name, mitdb_path);
fprintf('Loaded record %s at Fs=%d Hz\n', record_name, Fs);

%% 4. Bandpass Filter (0.5-40 Hz) with Guardrails
nyquist = Fs/2;
if nyquist <= 40
    error('Nyquist freq (%.1f Hz) ≤40 Hz. Increase Fs or reduce cutoff.', nyquist);
end

low_norm = max(0.5/nyquist, 0.001);  % Ensure >0
high_norm = min(40/nyquist, 0.999);  % Ensure <1

[b, a] = butter(3, [low_norm high_norm], 'bandpass');
ecg_filtered = filtfilt(b, a, ecg);

%% 5. Wavelet Analysis
% Parameters
wavelet_name = 'sym4';  % Symlet 4 is often good for ECG
level = 5;              % Decomposition level
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% Perform wavelet decomposition
[c, l] = wavedec(ecg_filtered, level, wavelet_name);

% Extract detail coefficients
d1 = detcoef(c, l, 1); % Level 1 details (highest frequency)
d2 = detcoef(c, l, 2); % Level 2 details
d3 = detcoef(c, l, 3); % Level 3 details
d4 = detcoef(c, l, 4); % Level 4 details
d5 = detcoef(c, l, 5); % Level 5 details

% Extract approximation coefficients
a5 = appcoef(c, l, wavelet_name, level);

% Reconstruct single branch signals
a5_rec = wrcoef('a', c, l, wavelet_name, level);
d5_rec = wrcoef('d', c, l, wavelet_name, 5);
d4_rec = wrcoef('d', c, l, wavelet_name, 4);
d3_rec = wrcoef('d', c, l, wavelet_name, 3);
d2_rec = wrcoef('d', c, l, wavelet_name, 2);
d1_rec = wrcoef('d', c, l, wavelet_name, 1);

%% 6. Visualization
t = (0:length(ecg)-1)/Fs;

colors = struct(...
    'original', [0, 0.4470, 0.7410],...
    'filtered', [0.8500, 0.3250, 0.0980],...
    'a5', [0.9290, 0.6940, 0.1250],...
    'd5', [0.4940, 0.1840, 0.5560],...
    'd4', [0.4660, 0.6740, 0.1880],...
    'd3', [0.3010, 0.7450, 0.9330],...
    'd2', [0.6350, 0.0780, 0.1840],...
    'd1', [0.5, 0.5, 0.5]...
);

figure('Position', [100, 100, 1200, 800]);

% Original and filtered ECG
subplot(4, 2, 1);
plot(t, ecg, 'Color', colors.original, 'LineWidth', 1.5);
title('Original ECG Signal', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 2);
plot(t, ecg_filtered, 'Color', colors.filtered, 'LineWidth', 1.5);
title('Bandpass Filtered ECG (0.5-40 Hz)', 'FontSize', 10);
xlabel('Time (s)');
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ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

% Wavelet decomposition components
subplot(4, 2, 3);
plot(t, a5_rec, 'Color', colors.a5, 'LineWidth', 1.5);
title('Approximation A5', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 4);
plot(t, d5_rec, 'Color', colors.d5, 'LineWidth', 1.5);
title('Detail D5', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 5);
plot(t, d4_rec, 'Color', colors.d4, 'LineWidth', 1.5);
title('Detail D4', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 6);
plot(t, d3_rec, 'Color', colors.d3, 'LineWidth', 1.5);
title('Detail D3', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 7);
plot(t, d2_rec, 'Color', colors.d2, 'LineWidth', 1.5);
title('Detail D2', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
set(gca, 'FontSize', 8);

subplot(4, 2, 8);
plot(t, d1_rec, 'Color', colors.d1, 'LineWidth', 1.5);
title('Detail D1', 'FontSize', 10);
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;
set(gca, 'FontSize', 8);

%% 7. Feature Extraction Plot
% Combine relevant detail levels for QRS detection
qrs_components = d3_rec + d4_rec + d5_rec;

% Simple threshold-based peak detection
thresh = 0.5 * max(abs(qrs_components));
[peaks, locs] = findpeaks(abs(qrs_components), 'MinPeakHeight', thresh,
'MinPeakDistance', Fs*0.2);

figure('Position', [100, 100, 1200, 600]);

% Plot the filtered ECG
subplot(2,1,1);
plot(t, ecg_filtered, 'Color', colors.filtered, 'LineWidth', 1.5);
hold on;
plot(locs/Fs, ecg_filtered(locs), 'o', ...
    'MarkerSize', 8, ...
    'MarkerEdgeColor', 'k', ...
    'MarkerFaceColor', [0.9 0.1 0.1]); % Red filled circles
title('ECG with Detected R-peaks', 'FontSize', 12);
xlabel('Time (s)', 'FontSize', 10);
ylabel('Amplitude', 'FontSize', 10);
legend('Filtered ECG', 'Detected R-peaks', 'Location', 'northeast');
grid on;
set(gca, 'FontSize', 9);

% Plot the QRS components used for detection
subplot(2,1,2);
plot(t, d3_rec, 'Color', colors.d3, 'LineWidth', 1.5);
hold on;
plot(t, d4_rec, 'Color', colors.d4, 'LineWidth', 1.5);
plot(t, d5_rec, 'Color', colors.d5, 'LineWidth', 1.5);
plot(t, qrs_components, 'Color', [0 0 0], 'LineWidth', 2, 'LineStyle', '--');
title('Wavelet Components Used for QRS Detection', 'FontSize', 12);
xlabel('Time (s)', 'FontSize', 10);
ylabel('Amplitude', 'FontSize', 10);
legend('D3', 'D4', 'D5', 'Combined (D3+D4+D5)', 'Location', 'northeast');
grid on;
set(gca, 'FontSize', 9);
```